Gauri Kshirsagar

Leicester | gaurikshirsagar51@gmail.com | +447352679460 | GitHub | LinkedIn

EDUCATION

SPPU: Bachelor of Engineering (B.E.) in Artificial Intelligence and Data Science. | Dec 2021 -Jun 2025

Key Modules: Artificial Intelligence, Deep Learning, Data Structures, Database Management Systems. (Maharashtra, India)

University of Leicester: MSc in Data science | Sep2025 -2026

Relevant Coursework: Statistical Modelling, Machine Learning, Data Visualization, Applied Data Analytics (Leicester, United Kingdom)

WORK EXPERIENCE.

Data Analytics Intern

IBM Skill Build (CSRBOX) | 15 Jul 2024 – 5 Aug 2024

- Cleaned and pre-processed structured datasets using data wrangling techniques to improve analysis accuracy.
- Conducted exploratory data analysis (EDA) and generated visual insights to identify trends and correlations.
- Presented key analytical findings to support data-driven decision-making and reporting processes.

Cybersecurity Intern

AICTE | Jul 2024 – Sep 2024

- Conducted vulnerability assessments across 5+ simulated lab environments, analysing common attack vectors including phishing, malware, and access misconfiguration.
- Identified and documented 20+ security vulnerabilities, improving threat detection accuracy and risk visibility through structured reporting.
- Facilitated implementation of mitigation strategies aligned with cybersecurity best practices, enhancing compliance awareness and incident response readiness

PROJECTS:

SPPU | Aug 2024-Aug2025

Subjective Answer Evaluation System | Python, NLP, Machine Learning

- Achieved **85% semantic similarity accuracy** across **850+ student responses**, reducing manual grading time by **40%** through NLP-based automated answer evaluation.
- Engineered TF-IDF and word embedding features and evaluated **5 machine learning algorithms** (Logistic Regression, SVM, Random Forest, KNN, Naïve Bayes), selecting the optimal model using **F1-score, accuracy, and cross-validation**.

UNIVERSITY OF LEICESTER | Oct 2025- Nov2025

European GHG Emissions Analysis | *R, Data Visualization, EDA*

- Cleaned Eurostat emissions dataset covering **2014–2023**, analysing **32 countries** and **12 key indicators**; resolved missing values, standardised variables, and improved data consistency for statistical modelling.
- Developed **9 analytical visualisations** (LOESS trend lines, heatmaps, density plots) to identify top-emitting regions, sectoral patterns, and 10-year emission trends across Europe.

Fraud Detection System (FinTech) | Python, Machine Learning | Feb2026 – Present

Ongoing Project

- Developed a fraud classification model on **75,000+ transactions** with **28 engineered features**, optimised using **precision, recall, F1-score and ROC-AUC (0.91+)**, applying **SMOTE and class weighting** for imbalanced data.
- Designed a **3-page Power BI dashboard** with **8 KPIs** (risk score, fraud rate, alerts, false positives, recall, precision, volume, latency) to support fraud investigation and risk monitoring.

CERTIFICATIONS:

Nashik (Maharashtra), India

- International Conference Paper Presentation – ICIA-MET'25 (April 2025)
- Data Science, ML, DL & NLP Bootcamp – Udemy (99 Hours, March 2025)

TECHNICAL SKILLS:

- **Programming:** Python (pandas, NumPy, scikit-learn, matplotlib), R (tidyverse, ggplot2)
- **Machine Learning:** Regression, Classification, Model Evaluation, Feature Engineering, Cross-Validation, Imbalanced Data Handling
- **Concepts:** NLP, TF-IDF, Embeddings, LOESS, Heatmaps, Fraud Detection.
- **Data Analysis:** Exploratory Data Analysis (EDA), Statistical Analysis, Data Cleaning, Data Wrangling
- **Visualization Tools:** Power BI, Tableau
- **Databases:** SQL (DDL, DML, DCL), MySQL